

M.Tech Mini Project — Climate Forecasting

Context & Objective

You are data scientists in a climate forecasting company. Your objective is to build, evaluate, and interpret time-series forecasting models to predict air temperature (T in °C) as accurately as possible using the Jena Climate dataset (10-minute sampling, Jan 10, 2009 — Dec 31, 2016). Focus on model accuracy, robustness, interpretability, and reproducibility.

Dataset Summary

Location: Weather Station, Max Planck Institute for Biogeochemistry, Jena, Germany

Timeframe: Jan 10, 2009 — Dec 31, 2016

Sampling frequency: 10 minutes

Feature Name	Description	Units	Type
Date Time	Timestamp of observation (local time)	-	Datetime
p (mbar)	Air pressure at the station level	Millibar (mbar)	Float
T (degC)	Air temperature at 2 meters above ground	°C	Float
Tpot (K)	Potential temperature (temperature adjusted for pressure)	Kelvin (K)	Float
Tdew (degC)	Dew point temperature	°C	Float

	(temperature of air saturation)		
rh (%)	Relative humidity	Percent (%)	Float
VPmax (mbar)	Maximum vapor pressure	Millibar (mbar)	Float
VPact (mbar)	Actual vapor pressure	Millibar (mbar)	Float
VPdef (mbar)	Vapor pressure deficit	Millibar (mbar)	Float
sh (g/kg)	Specific humidity (mass of water vapor per kg of air)	g/kg	Float
H2OC (mmol/mol)	Water vapor concentration	mmol/mol	Float
rho (g/m**3)	Air density	g/m ³	Float
wv (m/s)	Wind speed	m/s	Float
max. wv (m/s)	Maximum wind speed during the observation period	m/s	Float
wd (deg)	Wind direction	Degrees (°)	Float

Part A — Data Understanding & EDA (5 marks)

1. Data cleaning: handle missing data, wrong formats, timestamp parsing, and resampling (if used). Document choices and rationale.
2. Perform EDA: visualizations of temperature time series (daily/weekly/yearly), distribution plots, autocorrelation (ACF/PACF), and decomposition (additive & multiplicative). Comment on stationarity, seasonality, and trends.

3. Provide at least two visualizations that reveal non-obvious structure (e.g., diurnal cycles, seasonal patterns, correlation with humidity/pressure).

Part B — Baseline Models & Smoothing (5 marks)

1. Implement simple baselines:
 - Naive (last observed)
 - Seasonal Naive (same time previous day/week)
 - Simple Moving Average (window choices: 6, 24, 144 — justify)
2. Implement Exponential Smoothing variants: SES, Holt, Holt-Winters (additive and multiplicative where applicable).
3. Compare baselines using appropriate metrics on a hold-out test set.

Part C — Stationary Models (5 marks)

1. Test for stationarity (ADF) and apply differencing where necessary.
2. Fit AR, MA, ARMA models (use AIC/BIC for order selection). Show residual diagnostics (white noise, Ljung-Box test).
3. Forecast and compare with baselines. Provide interpretation of model coefficients and limitations for this dataset.

Part D — Non-Stationary & Seasonal Models (5 marks)

1. Build ARIMA and SARIMA models. Justify order $(p,d,q)(P,D,Q,m)$ selection using AIC and diagnostics.
2. Provide grid-search (or automated pmdarima/auto_arima) results and compare top candidates.

3. Produce multi-step forecasts (short horizon: next 3 hours; medium: next 24 hours; long: next 7 days) from each model and evaluate.

Part E — (Explore on your own and try complete this part)

Multivariate & Deep Learning Models (5 marks)

1. Build VAR/VARMA models using a subset of correlated features
2. Build at least two deep learning models for sequence forecasting (e.g., LSTM, GRU). Use appropriate input windowing (e.g., last 6–144 timesteps), normalization, and training/validation splits
3. Optionally, implement a Transformer-based time series model or Temporal Convolutional Network (TCN) for extra credit.
4. Compare performance with previous classical models and include computational cost (training time, inference time).